Big Data Analytics

Assignment – II

Topic 1: Decision Tree Construction using Entropy & Information Gain

Problem 1

Given the following dataset of weather conditions and play decisions, construct a decision tree using ID3 algorithm. Calculate entropy and information gain for each attribute.

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Problem 2

Using the dataset below about loan approval, apply the ID3 algorithm. Calculate entropy and information gain for each attribute and identify the root node.

ID	Credit Score	Income	Employment	Loan Approved
1	Good	High	Yes	Yes
2	Good	High	No	Yes
3	Fair	High	Yes	Yes
4	Poor	Medium	Yes	No
5	Poor	Low	Yes	No
6	Poor	Low	No	No
7	Fair	Low	No	Yes
8	Good	Medium	Yes	Yes
9	Good	Low	Yes	No
10	Poor	Medium	No	No
11	Good	Medium	No	Yes
12	Fair	Medium	Yes	Yes
13	Fair	High	No	Yes
14	Poor	Medium	Yes	No

Topic 2: Association Rule Generation using Apriori Algorithm

Problem 3

A supermarket has the following transaction database. Apply the Apriori algorithm with min_support = 50% and min_confidence = 70%. Find all frequent itemsets and generate strong association rules. Calculate Lift for each rule.

TID	Items Purchased
T1	Bread, Milk, Eggs
T2	Bread, Butter
T3	Milk, Butter, Cheese
T4	Bread, Milk, Butter
T5	Bread, Milk, Eggs, Butter
T6	Milk, Eggs
T7	Bread, Eggs
T8	Milk, Butter

Problem 4

Apply the Apriori algorithm on the following bookstore transaction data with min_support = 40% and min_confidence = 60%. Generate all association rules and compute Support, Confidence, and Lift.

TID	Books Purchased
T1	Python, ML, Statistics
T2	Python, DataScience
T3	ML, Statistics, DataScience
T4	Python, ML
T5	Python, ML, DataScience

Topic 3: User-Based Collaborative Filtering

Problem 5

The following matrix shows movie ratings (1-5) given by users. Generate recommendations for User 5 using User-Based Collaborative Filtering. Use Pearson Correlation to find similar users. Predict the rating for Movie 4.

User	Movie1	Movie2	Movie3	Movie4	Movie5
User1	5	3	4	4	2
User2	3	1	2	3	3
User3	4	3	4	3	5
User4	3	3	1	5	4
User5	1	5	5	?	2

Problem 6

Using the book ratings table below, apply User-Based CF with cosine similarity to predict what rating User A would give to Book 4. Use the top-2 most similar users.

User	Book1	Book2	Book3	Book4
User A	4	0	5	?
User B	5	4	4	2
User C	0	0	0	5
User D	2	4	0	4
User E	3	3	4	3

Topic 4: Item-Based Collaborative Filtering

Problem 7

Using the following user-item rating matrix, apply Item-Based Collaborative Filtering with cosine similarity to predict the rating User C would give to Item 3.

User	Item1	Item2	Item3	Item4	Item5
User A	4	3	0	5	1
User B	5	0	4	4	2
User C	3	1	?	3	2
User D	0	0	4	0	0
User E	1	0	5	2	0

Problem 8

Apply Item-Based CF using adjusted cosine similarity on the following rating data. Predict the ratings of User 2 and User 5 would give to Product 5.

User	P1	P2	P3	P4	P5
User1	3	4	3	1	2
User2	4	3	4	1	?
User3	3	3	1	2	4
User4	1	1	2	3	3
User5	2	1	5	1	?

Topic 5: Content-Based Recommendation

Problem 9

The following table shows movies with their content features. User A has watched and liked Action=High, Sci-Fi=High, Romance=Low, Comedy=Low. Using content-based filtering with cosine similarity, recommend the top 2 movies for User A from the unwatched list.

Movie	Action	Sci-Fi	Romance	Comedy	Watched by A?
Movie1	5	5	1	1	Yes (liked)
Movie2	4	4	2	1	No
Movie3	1	1	5	4	No
Movie4	5	3	1	2	No
Movie5	2	2	4	5	No
Movie6	4	5	1	1	No

Problem 10

Using TF-IDF based content-based filtering, recommend articles to a user who liked Article 1. The articles have the following keyword presence (1=present, 0=absent). Use cosine similarity to find the most similar article.

Article	ML	AI	Python	Java	BigData	Cloud	Stats
Article1	1	1	1	0	1	0	1
Article2	1	1	0	0	1	0	1
Article3	0	0	0	1	0	1	0
Article4	1	0	1	0	0	0	1
Article5	0	1	0	0	1	1	0

Apply TF-IDF weights: TF=1 (if keyword present), IDF = log(N/df) where N=5 articles, df=document frequency.